FOUNDATION FOR FOOD & AGRICULTURE RESEARCH

FFAR Final Progress Report

As part of closing out a FFAR grant, grantees must complete a Final Progress Report. Please use the template below to complete the programmatic report. The report must be submitted to FFAR within 90 days after the expiration or termination of a FFAR funded research grant. All questions about this form should be directed to grants@foundationfar.org.

The Final Progress Report communicates the cumulative results and major accomplishments of the funded grant research, including accomplishment for each aim and whether all research aims were fully completed. FFAR uses the content of the Final Progress Report to inform Congress, the Advisory Board, and other stakeholders on the successes, impacts and value of funding unique partnerships to support innovative science addressing today's food and agriculture challenges. If a question is not applicable to the project, please type "Nothing to report."

Grant Information	
Grant ID	DSnew-0000000027
Award Program	
Project Title	Enabling Data Sharing Between Farmers and Agricultural Scientists through a Research Database
Grant Start Date	01/14/2021
Grant End Date	01/14/2022
Total Grant Budget	$105,555.56
Total FFAR Budget	$51,666.67
Matching Funder(s)	Field to Market Walton Family Foundation Edge Dairy Cooperative
Final Report Date	04/14/2022
Project Director/Principal Investigator Information	
Full Name	Jamie Richards (originally Allison Thomson)
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Grantee Organization Information	
Organization Name	Field to Market: The Alliance for Sustainable Agriculture
Address	777 N Capitol St NE Suite 802
City, State Zip	Washington DC 20002
Tax ID	90-0885216
Authorized Signing Official (ASO) Information	

Name	W. Scott Herndon
ASO Title	President
Email	Sherndon@fieldtomarket.org
Phone	202-470-5583

General Information

1.1. **Please list the geographic location(s) – city, state, congressional district – where the work was conducted. If the work was conducted outside of the US, please list the city and country.**

Washington, DC.

1.2. **How many new jobs were created because of this grant funding?**

None.

1.3. **How many jobs were maintained because of this grant funding?**

¼ position was maintained with funding

1.4. **Have there been any changes to your organization's IRS 501(c)(3) non-profitstatus since you were awarded the grant? If yes, please explain**.

No

1.5. **Has your organization undergone a recent name change? If so, please providethe new name of your organization.**

No

2. Scientific Report

2.1. **Public Abstract (up to 500 words): The abstract should be 'stand-alone' and is intended for a general audience. It should be a concise**

overview/summary of theimportance of the project, the issues that the project addressed, and the key findings of the project. The abstract should also clearly state how the results of the project help to address important needs in U.S. food and agriculture systems.

Researchers rely on a combination of public statistics, remote sensing and data shared by production farms to advance our understanding of how farm management impacts the environment. Farmers rely on this research, translated into specific recommendations by agronomic advisors, when making management decisions to meet both production and environmental stewardship goals such as improving soil health and reducing greenhouse gas emissions.

Complete records from production farms are difficult to obtain, so researchers typically depend on data collected on research farms, which may not reflect actual conditions in commercial operations. While many decision support tools collect a wealth of on-farm data, that prove invaluable for research, data privacy and corporate policies restrict access, limiting research opportunities for better on-farm sustainability.

Field to Market has the Fieldprint Platform, which collects and analyzes data from over 4,000 producers farming over 4.5 million acres every year. To improve researcher access to actual farm production data, Field to Market is making data entered into Fieldprint Platform available in aggregated and anonymized form with the express permission of the farmer. With support from FFAR, Field to Market embarked on a research project to i) implement an "opt-in" feature on the Fieldprint Platform for farmers to share their data, ii) identify any legal and non-legal barriers to implementation of such a feature, and iii) develop the technological infrastructure to populate the research database with data from both the free, online Fieldprint Calculator and Qualified Data Management Partners (QDMPs), which are farm management programs that have embedded the analytics of the Fieldprint Platform.

When a farmer opts-in to the Research Database, the Fieldprint Platform (either as the Fieldprint Calculator or through a QDMP) flags the farmer's data for inclusion in a research database catalog, which is stripped of any personal information.

The technology behind the "opt-in" feature and associated infrastructure to aggregate and produce catalogs was developed successfully under the FFAR grant. A Standard Operating Procedure (SOP) for Governance of the Research Database was developed that will be used by the Field to Market Science Advisory

Council to determine if a request for use of a catalogue will be granted. At the time of this report, the opt-in feature is available to users of the free, online Calculator and approximately 40 actual farm records have been collected for research purposes. Field to Market will publish the first catalogue of data after sufficient data becomes available for research.

This effort to increase access to anonymized data from production farms will help scientists better characterize existing farm management practices and provide insights back to farmers.

2.2. **Provide a description or interpretation of how the key findings of the project can be used by the agriculture community. It is important the PI identifies how these findings have been, or may be, adopted or adapted in U.S. agriculture and food systems. If the findings cannot yet be applied, this section should address how they can be used to guide future planning or decision-making (up to 1,000-wordlimit).**

This project developed a novel mechanism for farmers to share their field data with scientists and will benefit the research community by expanding the sources of information available to them on conducting research around adoption of practices and the effects of those practices on environmental metrics. In particular, the project will make available anonymized field-level farm management records and Field to Market's environmental metric calculations for those farmers who opt-in. The full impact of that data availability will depend on the volume of data collected.

The mechanism developed for data sharing and the surrounding implementation approach and decisions were developed in coordination with Field to Market's Metrics Standing Committee and Science Advisory Council, discussed at various membership meetings attended by a diverse set of supply chain stakeholders, including grower sector organizations, and is based on feedback and input gathered from discussions with seven of Field to Market's qualified data management partners (QDMPs). The QDMPs include several of the largest and leading precision agriculture software platforms. This helped ensure the mechanism and approach to implementation, which the research database was built upon was vetted among the very users contributing to, implementing, supporting, or using the research database. Several specific findings helpful to the agriculture community arose from the work.

Establishing Governance: First, a model request and approval process to govern access to a research database was developed. Among other areas, the model – or Standard Operating Procedure for Governance of the Research Database – establishes the process for management of and access to the database and how research results can be disseminated and used. The intent was to instill confidence in the overall governance and use of the research database. The SOP developed can serve as a blueprint for governance of similar projects.

Identifying Barriers: Another area focused on the extent to which legal or other barriers existed, preventing or limiting interest in Field to Market's QDMPs implementing the research database concept within their own platforms. Over 80 percent of the data processed through the Fieldprint Platform is stored within QDMPs systems. Therefore, identifying barriers to implementation was critical to understanding if the research databases can be effectively scaled and thus provide (over time) a meaningful volume of data for research. Findings indicate there are no strong objections or specific legal barriers from among the seven QDMPs interviewed. Knowing that Field to Market's QDMPs – most being, leading precision agriculture providers in the U.S. – did not identify material barriers to participation in a scientific research database sets an important precedence within the agricultural community.

Effective Communication: Recognizing that farmers are often reluctant to share data due to privacy or other concerns, another key finding area was to better understand how the research database concept should be positioned or communicated to farmers. Through discussions with membership at large, engaging with various committees, and holding discussions with QDMPs, Field to Market learned how to best communicate the value or benefit of the concept more effectively to both farmers and agronomic advisors. Specific communication materials were developed based on these findings. The learnings from these findings and the materials collected can serve as examples for similar endeavors.

Balancing Data Privacy: Finally, with the ability to work directly with the scientific and grower communities engaged in Field to Market, findings from this work included developing a research catalog data model that aimed to balance the need by researchers to access a robust data set with the very real data privacy concerns commonly expressed by farmers. From the researcher's perspective, the catalog data model is based on individual crop year or growing season records (e.g. 2020 Corn on Field A). The records are robust and based on the input data and output data required as part of Field to Market's sustainability metric framework. At the same time, to ensure data privacy, data fields such as field boundary coordinates and farmer or field names are removed prior to generating the catalog researchers would access.

2.3. **Research Outcomes/Impact (up to 1,500-word limit). The primary goal here is toanswer questions such as "How did this project lead to improvements in U.S. agriculture and food systems?" or "How can the findings of this study guide or make improvement in future investigations and research?" Outcomes should be explained and classified in one of the following ways:**

a) **potential outcomes, i.e., findings, results, or recommendations that couldimpact U.S. agriculture and food systems, if used;**
b) **b. intermediate outcomes, i.e., how the findings, results, or recommendations have been used by others to influence practices,legislation, research/product design, training and so forth; and**
c) **c. end outcomes, i.e., how findings, results, or recommendations havecontributed to documented reductions in work-related morbidity, mortality, and/or exposure.**

Building off the key finding areas outlined in Section 2.2, this section expands on how outcomes of the project can directly guide or make improvements in future investigations and research in the agricultural community.

Establishing Governance: Development of the Standard Operating Procedure for Governance of the Research Database provides a model request and approval process other researchers can use or adapt for similar research database efforts.

Identifying Barriers: That no legal or other material barriers to implementation were identified based on discussions with Field to Market's seven QDMPs, offers other researchers some level of assurance that similar projects can partner with industry players without encountering roadblocks of this nature.

Effective Communication: Future research efforts can leverage the approaches to communication developed under this work. As farmer participation is critical to the success of efforts to collect research data, having a starting place for how best to communicate the value and instill confidence in the process is important.

Balancing Data Privacy: Perhaps key to supporting future research is the time and effort spent in discussing and finalizing an approach to balance data needed by researchers with the privacy concerns commonly raised by farmers. The data model for data collection and aggregation can be leveraged by future projects.

2.4. **What were the goals/specific aims of the project? If the approved application listsmilestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets. (up to 1,500-word limit)**

Within the application there were three main goals for the project:

Goal 1: Development of programming code and logic to capture and store data securely and anonymously and make it available to researchers and implement the process within Field to Market's Fieldprint Platform.

Goal 2: Goal two included two subcomponents. First was outreach to farmers and commodity organization representatives to provide feedback on the technology implementation and data privacy protocols. Second was to explore and resolve any legal barriers discovered to participate in the program through discussions with Field to Market's QDMPs. The discussions for this part of the project took place from spring of 2021 through the end of 2021.

Goal 3: Populate the Research Database by promotion to farmer participants. Promotion of the project took place throughout the entire length of the project at various stages. The research database was promoted at events such as Field to Market's Summer 2021 and Fall 2021 plenary meetings and the Commodity Classic event in Winter 2022. Populating the database started in fall 2021 and will continue beyond this completion of this project.

All goals identified in the application have been completed at the time of this final report.

2.5. **Have any of the major goals/specific aims or milestones for the project changedsince the award? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals. (up to 500-word limit)**

Goal 2 – to explore and resolve legal barriers – was adjusted partway through the project. After initial conversations with the seven private sector data partners, it became clear that the barriers to their participation were not legal. Therefore, we pivoted the project to spend that time instead developing communications material, hosting a farmer focus group, and testing communications strategies with the data partners. The overall goal was the same – remove barriers to data partner participation – but the approach and tasks were different than in the original proposal.

2.6. **What was accomplished under the goals/specific aims or milestones for the project? Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (bothpositive and negative); and 4) key outcomes or other achievements. Include discussion of stated goals not met. The emphasis in reporting in this section should focus on accomplishments. In the response, emphasize the significance of the findings to the scientific field. This section can be as technical as the PI would like. (up to 5,000-word limit)**

Within the application there were three main goals identified for the project. This response focuses on what was accomplished under each of the three goals and then summarize the overall findings of the project.

Goal 1 for the project was the development of technology infrastructure to capture, store and manage the information required for the Research Database. This major activity was executed as planned with no major obstacles encountered.

Research Database Model: Initial work focused on the research database data model, balancing the data needed by researchers with the privacy concerns commonly expressed by farmers. For example, to ensure compliance with data privacy agreements, it was necessary to develop a meta-data system that would include details on weather stations and soils datasets without conveying exact field location details. Soils datasets are relatively standard in the research and modeling communities and identifying and communicating the soils identifier in the database infrastructure went smoothly. However, in exploring what information on weather stations to include, the project encountered some challenges specific to how USDA models in the Platform acquire wind station data. By working with USDA model developers and science advisors, we developed a system to capture the necessary station identifiers so that users of the research database would be able to accurately make use of the farm management information with the correct environmental information. Other decisions were made to protect data privacy such as removing field boundary coordinates and anonymizing field names prior to adding the data to the research database. The Research Database page in the Platform includes a copy of the research database data model so that users can see specifically what data is included.

Standard Operating Procedure: Another key area involved development of a Standard Operating Procedure for Governance of the Research Database (SOP). The SOP was developed at the beginning of the project and vetted through numerous Field to Market members and stakeholders. The document was then announced and published on the Field to Market member portal (now Version 1.1). The SOP describes the "opt-in" process, management of the database, granting access to the research database, dissemination and use of the research results, and the various forms used in the process (e.g. review, approval and agreement forms). Elements of language in the SOP were then adapted for use on the Research Database webpage within the Fieldprint Platform Calculator.

Fieldprint Platform Opt-In / Research Database Page: First, the Opt-In feature and a dedicated research database page was added to the Calculator. The additions were released in the Platform in the fall of 2021. When a farmer logs into the Calculator, they are prompted with an informational dialog box describing the research database and offering the option to either opt-in to the research database, learn more about the research database, dismiss the opt-in action for their current visit, or dismiss participation altogether. The new research database page (also available from the farmers profile menu) includes the farmer's current opt in status alongside information about the research database including benefits, data privacy and FAQs.

Second, a secure, dedicated Field to Market staff administration page was developed for managing the research database catalogs. The page allows staff to view and download research database catalogs and upload yearly data submissions from QDMPs. As an alternative to emailing submissions to Field to Market staff, a dedicated page was created that allows QDMPs to upload data directly to the database. Prior to generating the yearly research database catalog, any QDMP submitted data is preprocessed. This preprocessing includes recalculating data on the latest metrics version (if not current), inserting weather station data (if absent), and removing specific identifying information (e.g. field boundaries). Finally, the research database processing combines data from the Calculator with QDMPs data and the research database catalog is saved.

The administrative page functionality was released to the Calculator in the first quarter of 2022. This backend Platform infrastructure provides a solid framework to grow and handle future participation in the research database without any further development required. We also developed an implementation guide for QDMPs that provides an overview of the project, details on the data model, and guidance information they can use in their own software platforms.

Conclusion: The development of technology infrastructure to capture, store and manage the information required for the Research Database was successful. The infrastructure is in place to scale the research database in a way that provides researchers with needed data, satisfies farmer data privacy, and establishes a governance process for management and use. To validate our technical work, we engaged one QDMP to provide feedback on the overall technical approach, discuss the data model with respect to data privacy, discuss the approach to submitting QDMP data to Field to Market, and provide input on the QDMP research database implementation guide developed under this work.

Goal 2 was to explore and resolve any legal barriers discovered to participate in the program through discussions with Field to Market's Qualified Data Management Partners (QDMPs). A major activity was to work with the private sector data partners (QDMPs) to identify any concerns or barriers and encourage participation by making the farmer opt-in available to users of their platforms. QDMPs were introduced to the project as part of routine quarterly QDMP meetings. We followed up individually with each QDMP for discussion based on the following interview questions:

- What benefits do you see from the Research Database?
- What risks do you see for your organization and/or for your farmer customers?
- Do your current data agreements with farmers allow for participation?
    - If not, can you describe what provisions would prohibit it?
- Are there other legal or business barriers to your participation?

A key finding from QDMP interviews was that all had similar user license agreements with their farmers. All provisions for sharing data with third parties so long as there is explicit permission from the farmer to do so (e.g. opt-in). The wording of those agreements therefore would not be a barrier to supporting the research database within their platforms.

All QDMPs agreed the concept of a research database would offer value to the agriculture and scientific community at large. However, they did emphasize the need to clearly demonstrate the value and benefit to farmers. They noted potential relationship management or reputation concerns. More generally, they indicated they would exercise caution in how they communicated with farmers any new option for third party data sharing. This stems from the sense that farmers are becoming more aware that their data has increasing value.

From another dimension, responses to questions varied slightly depending on the size and structure of the QDMP organization. While the addition of an opt-in feature is seemingly simple, larger QDMPs and QDMPs with national retailer networks expressed concern around the time and effort required to introduce a new feature within their software products and to train staff on how to message and respond to questions. Smaller QDMPs with more direct farmer relationships focused more on the cost of implementation. Assuming no barriers from farmers, all QDMPs indicated they would consider adding the research database if the customers they support indicated a demand.

Conclusion: A clearer understanding of the concerns held by QDMPs informed development of the research database infrastructure. Learning that legal factors were not likely to be a barrier to implementation was important. Also, learning the importance of clearly communicating value and benefits to farmers helped develop more targeted farmer communication materials. Lastly, recognizing that implementation considerations will differ based on QDMP size and organization will help Field to Market better support QDMPs when they choose to implement the research database.

Goal 3 focused on population of the Research Database by promotion to farmer participants. The research database project was socialized through a number of Field to Market's program opportunities including committee meetings (e.g. metrics), project administrators network communications, Board of Director meetings, the most recent Science Advisory Council meeting, and the most recent semi-annual Field to Market member meeting. These opportunities allowed for promotion of the research database and for gathering feedback and ideas. Field to Market staff also coordinated a farmer focus group to test out messaging and understand how the Research Database concept would be received.

The messaging developed under this work was clear and compelling to the farmers, and they indicated they would participate if offered the opportunity. Key findings from the communications development and farmer focus groups were:

1. to focus on the reputational benefit, trust and social license to operate

outcomes of participating in increasing the transparency of farm operations;

2. to emphasize that the research database would be for non-commercial use only and that no one would profit from their data; and
3. to provide a specific example to potential users of what data would be shared.
4. Finally, they shared an appreciation for the inclusion of an “opt-out” feature that allows famers to rescind their permission after a positive opt-in.

This feedback informed the design and development of the web interface and the wording of the opt-in request. It also informed the development of further communications materials for the QDMPs, which were presented to them as a group with individual follow-up. It will also be used for promotion of the database to farmer participants of Field to Market projects during the winter data entry season.

Overall, the project’s findings made clear that the burden for creating interest among growers for participating in the Research Database falls to Field to Market, potential data users and other partners. These findings can be used by others seeking to understand how to engage with farmers more productively in research projects and programs, as well as researchers seeking to partner with private sector farm data management systems. Understanding what the most effective communications strategies are up front can increase the chances of building successful partnerships. The ultimate product of the project will be the Research Database itself, and the impact of that for the research community will depend on the scale and scope of farmer participation. The goal of the database is to create a robust, multi-year look at fields in different locations across the country and identify if practice changes, implementation of best management practices or other on-farm changes have impacted sustainability goals. As previously stated, the impact the data will have on the agricultural community is based upon the interest in the farmer to “opt-in”. Field to Market expects it to take 2-5 years of promotion to get a substantial number of growers to participate in the research database. Future findings by researchers using the database will determine the value the data has to the research community. The database will be an ongoing feature of the Fieldprint Platform and is designed for participation to grow year over year.

The next steps for the project after this grant has ended is to continue to promote the research database to both growers engaged with Field to Market, either through the online tool or QDMPs. The promotion will also occur at upcoming Field to Market meetings. It is anticipated the first research database catalogue will become available for requests in Summer of 2023. There is also a possibility for continued research around answering the questions about grower and QDMP incentives to grow participation as well is the value the research database has to researchers and what else might be learned through research done with the database.

2.7. **Discuss efforts taken to ensure the approach is scientifically rigorous. Include approaches taken to ensure robust and unbiased results.**

When conducting QDMP and farmer interviews and focus groups we ensured a standard approach, providing all participants with the same information and asking a standard set of questions.

Regarding the research database data model, the work followed standards and tested database structures that are currently in use by the Fieldprint Platform. This ensures continuity between the inputted data from the individual farmer and the Research Database.

In terms of governance around the use of the research data, Field to Market will review all research proposals requesting access to the data. Data will not be provided unless the proposal meets the criteria being used to evaluate a proposal by the Science Advisory Council. In brief, the Science Advisory Council will consider:

1. Research question must be provided
2. Research funding source
3. Be related to data being provided from Research Database
4. Not related to proprietary or for-profit software (no commercial use)
5. Specified crop, region, data of interest
6. Commit to objective and fair use of data
7. Subject to Field to Market and Peer Review process to ensure compliance
8. Acknowledge Field to Market’s Research Database in publication
9. Not be verified claims from Field to Market
10. Researchers will become Research Members of Field to Market

A more extensive list and discussion of the above points can be found on the

Field to Market member website in a document titled “Standard Operating Procedure for Governance of the Research Database”.

Researchers must commit to using the information fairly and objectively. After the research has been completed, Field to Market will request to review the findings one month prior to publication and ensure that the data is used appropriately. To stay transparent about data use, Field to Market will list all approved research projects and publications from the Research Database on Field to Market’s website.

2.8. **List key stakeholders that could be served by results of this project.**

The primary stakeholder is the U.S. research community engaged in studies of the environmental impact of management practices on commodity crop fields who will directly benefit from the Research Database itself. Additional stakeholders include farmers and farmer organizations (such as matching funder Edge Dairy Farmer Cooperative and grower organizations for major commodity crops) as the database provides their members with an opportunity to share data, leading to greater transparency and helping advance research relevant to their crops.

2.9. **How have the results of this reporting period been disseminated to communities of interest? Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write “nothing to report.” A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sourceswill be reported under Information Products. (up to 1,000-word limit)**

The Research Database opt-in has been deployed on the Fieldprint Platform website complete with descriptions and a frequently asked questions information sheet. Notice of this has been provided to the Field to Market Project Administrators who serve as the interface between farmers and the Platform. Project Administrators can communicate the Research Database opportunity to farmers. Notice has also been provided to all Field to Market members through full member update documents and presentations. QDMPs have been contacted and all QDMPs have been provided presentation on the research database project along with a submittal guide and standard operating procedures for the research database program. Individual meetings have taken place between QDMPs and Field to Market to make sure all voices were heard, and issues discussed.

Announcements of the research database project were also provided as briefings to Field to Market members at committee meetings and two annual member meetings. The tool was also promoted at an exhibit booth at the 2022 Commodity Classic to promote the use to growers.

Further outreach and promotion to Field to Market members, farmers, and the public is planned for as a press release, the Field to Market semi-annual member meeting and QDMP follow-up meetings for 2022.

2.10. **Describe challenges or delays encountered during project and actions that weretaken to resolve them. Only describe significant challenges that may have impeded the research and emphasize their resolution. (up to 500-word limit)**

There were no significant delays. The major challenge was anticipated – the reluctance of the QDMPs to develop the opt-in feature in their platforms. We worked with them to understand the concerns and provide the needed information.

2.11. **What opportunities for training and professional development has the project provided during this reporting period? If the research is not intended to providetraining and professional development during this period, state "Nothing to Report." For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual DevelopmentPlans (IDPs) are used to help manage the training for those individuals. (up to500-word limit)**

Nothing to Report

2.12. **Please indicate the number of undergraduate and graduate students, post- doctoral scholars, or other educational components involved during this reportingperiod. If other education components are involved, please describe them in detail. (up to 500-word limit)**

None

3. Information Products

3.1. **Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video,audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during the project resulting directly from the FFAR award.**

The following information products were developed to support the research database.

- Standard Operating Procedure for Governance of the Research Database document. The SOP describes the "opt-in" process, management of the database, granting access to the research database, dissemination and use of the research results, and the various forms used in the process (e.g. review, approval and agreement forms). The document is stored in the Field to Market Learning Center (https://members.fieldtomarket.org/members/learning-center/sop-governance-research-database)
- Fieldprint Calculator Research Database Web Page: An informational web page available to registered users of the Fieldprint Calculator. The page is accessible through the user profile menu. The page includes the opt in status, provides an overview of the research database, and lists frequently asked questions. (https://calculator.fieldtomarket.org)
- Research Database Data Model Example: Linked to the Fieldprint Calculator Research Database web page and added for the benefit of farmers so they understand the type of data researchers will see, this

PDF document shows the research database data model (or fields) and sample data. (https://calculator.fieldtomarket.org)

- Research Database Communication Tips and FAQs. This document provides communication tips and frequently asked questions for QDMPs implementing the research database within their own software platforms. The document is stored in the Field to Market Learning Center (https://members.fieldtomarket.org/members/learning-center/sop-governance-research-database)
- Research Database QDMP Submittal Guide document: The guide provides QDMPs with information on how to implement the opt in feature within their platform, explains the approach to data privacy, and outlines the process for the yearly catalogs are processed. The document is stored in the Field to Market Learning Center (https://members.fieldtomarket.org/members/learning-center/sop-governance-research-database)

3.2. **Please provide a list of citations for the information products produced during the project.**

The main development product can be viewed at:
https://calculator.fieldtomarket.org (requires user account)

3.3. **Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during theproject resulting directly from the FFAR award? If yes, please provide citation.**

None

3.4. **Website(s). List the URL for any internet site(s) that disseminates the researchactivities. A short description of each site should be provided. It is not necessary to include the publications already specified above.**

See Section 3.2 of this report.

3.5. **Have inventions, patent applications, and/or licenses resulted from the project(U.S. and international)? If yes, indicate the invention, patent application(s), and/or license(s).**

None

3.6. **Are any of the information products produced during this grant that are confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if theymust remain confidential indefinitely. (up to 500-word limit)**

None

3.7. **Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers,and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that other can access andre-use these data in the future?**

As noted elsewhere in this report, the research database and the related products developed have been (and will continue to be) communicated throughout Field to Market's membership. QDMPs implementing the research database will further promote the project.

4. Data Management

4.1. **Did the project generate any data? Data generation includes transformation of existing data sets and data from existing resources (e.g., maps and imageries). Please list the data generated for the award.**

None

4.2. **If you list multiple data sets, are these data sets related? If so, please provide ashort description of how they are related.**

4.3. **Please provide copies of relevant metadata records to support FFAR's mission ofenhancing the discoverability of FFAR funded project data and information products. Include or attach copies of records and a simple file inventory, if necessary, in a compressed folder.**

FOUNDATION FOR FOOD & AGRICULTURE RESEARCH

Certification

The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate and complete. I am aware there is a significant penalty for submitting false or misleading information.

If applicable, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I willnot claim otherwise.

PI Full Name: Jamie Richards

PI Signature: ______________________ Date: April 11, 2022

Authorized Signing Official (ASO) Name: Scott Herndon

ASO Signature: Scott Herndon Date: April 13, 2022